

**SECOND SEMESTER (CUCBCSS—UG) DEGREE EXAMINATION
APRIL 2021**

Mathematics

MAT 2C 02—MATHEMATICS

Time : Three Hours

Maximum : 80 Marks

Part A (Objective Type Questions)

Answer all twelve questions.

Each question carries 1 mark.

1. Find $\int a^x dx$.
2. Define the partial derivative of $f(x, y)$ with respect to x at (x_0, y_0) .
3. If f is continuous on $[a, \infty)$, then $\lim_{b \rightarrow \infty} \int_a^b f(x) dx = \dots\dots\dots$
4. If a series $\sum a_n$ converges, then $\lim a_n = \dots\dots\dots$
5. Find the n^{th} term of the sequence $-2, 2, -2, 2, \dots\dots\dots$
6. Find the domain of the function $z = \sin xy$.
7. The polar form of the line $y = 2$ is $\dots\dots\dots$
8. $\frac{d}{dx} \cosh x = \dots\dots\dots$
9. If $f(x, y) = 2x^2y$ then find $\frac{\partial^2 f}{\partial x \partial y}$.
10. Give an example of conditionally converging series.

11. State Sandwich theorem for sequences.
12. Write the transformation equations for Cartesian co-ordinates to spherical polar co-ordinates.

(12 × 1 = 12 marks)

Part B (Short Answer Type)

Answer any **nine** questions.

Each question carries 2 marks.

13. Evaluate $\int_0^{\log 2} 4e^x \sinh x \, dx$.
14. The region between the curve $y = \sqrt{x}$, $0 \leq x \leq 4$ and the x -axis is revolved about the x -axis to generate a solid. Find its volume.
15. Determine whether the sequence $a_n = \frac{2n+1}{3n+1}$ is non-decreasing and if it is bounded from above.
16. Describe the level surface of the function $f(x, y, z) = \sqrt{x^2 + y^2 + z^2} - 1$.
17. The plane $x = 1$ intersects the paraboloid $z = x^2 + y^2$ in a parabola. Find the slope of the tangent to the parabola at $(1, 2, 5)$.
18. Find $\frac{\partial z}{\partial y}$ if the equation $yz - \ln z = x + y$ defines z as a function of the two independent variables x and y and the partial derivatives exists.
19. Is the area under the curve $y = \ln x/x^2$ from $x = 1$ to $x = \infty$ finite? If so, what is it?
20. Evaluate $\int_0^1 \frac{dx}{\sqrt{3+4x^2}}$.
21. Write the polar form of the circle $x^2 + (y-3)^2 = 9$.
22. Draw the polar curve $r = 2 \cos \theta$.

23. Find a spherical co-ordinate equation for the sphere $x^2 + y^2 + (z - 1)^2 = 1$.

24. Find the volume of $\sum_{n=1}^{\infty} \frac{1}{n(n+1)}$.

(9 × 2 = 18 marks)

Part C (Short Essay Type)

Answer any six questions.

Each question carries 5 marks.

25. Find the volume of the solid generated by revolving the region bounded by $y = \sqrt{x}$ and the lines $y = 1, x = 4$ about the line $y = 1$.

26. Investigate the convergence of $\int_0^3 \frac{dx}{(x-1)^{2/3}}$.

27. Does the sequence whose n th term is $\left(\frac{n+1}{n-1}\right)^n$ converge? If so, find its limit.

28. Find the sum of the series $\sum_{n=1}^{\infty} \frac{2^{n-1} - 1}{3^n}$.

29. Find the linearization of $f(x, y, z) = x^2 - xy + 3 \sin z$ at the point $(2, 1, 0)$.

30. Express $\frac{\partial w}{\partial r}$ and $\frac{\partial w}{\partial s}$ in terms of r and s if $w = 2x + 2y - z^2, x = r/s, y = r^2 + \ln s, z = 2r$.

31. Find the length of the curve $x = \frac{y^3}{3} + \frac{1}{4y}$ from $y = 1$ to $y = 3$.

32. Show that $\sum_{n=1}^{\infty} \frac{\ln n}{n^{3/2}}$ converges.

33. Find the radius and interval of convergence of the series $\sum_{n=0}^{\infty} (-1)^n (4x+1)^n$.

(6 × 5 = 30 marks)

Part D (Essay Type)

Answer any two questions.

Each question carries 10 marks.

34. Find the length of the cardioid $r = 1 - \cos \theta$.

35. a) Evaluate $\int_{-\infty}^{\infty} \frac{dx}{1+x^2}$.

b) Show that $\tanh^2 x = 1 - \operatorname{sech}^2 x$

36. a) Using partial differentiation find $w'(0)$ if $w = xy + z$, $x = \cos t$, $y = \sin t$, $z = t$.

b) If $f(x-y, y-z, z-x) = 0$, show that $\frac{\partial f}{\partial x} + \frac{\partial f}{\partial y} + \frac{\partial f}{\partial z} = 0$.

(2 × 10 = 20 marks)